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 Studierichting: Informatica
 Oefeningenreeks 3H nr. 9.2

$$r = 2 - \cos \theta$$

Domein: $\text{dom } r = \mathbb{R}$, beperkt domein: $[0, 2\pi]$

Periode: $P = 2\pi$

Nulpunten:

$$r = 0 \Leftrightarrow \cos \theta = 2 \Leftrightarrow \text{geen oplossingen } (\emptyset)$$

Afgeleide en extrema:

$$r' = \sin \theta = 0 \Leftrightarrow \theta = k\pi \rightarrow \{0, \pi, 2\pi\}$$

Verloopstabel:

θ	0		π		2π
r	1	+	3	+	1
r'	0	+	0	-	0
	min	$\nearrow$	max	$\searrow$	min
	1		3		1

References

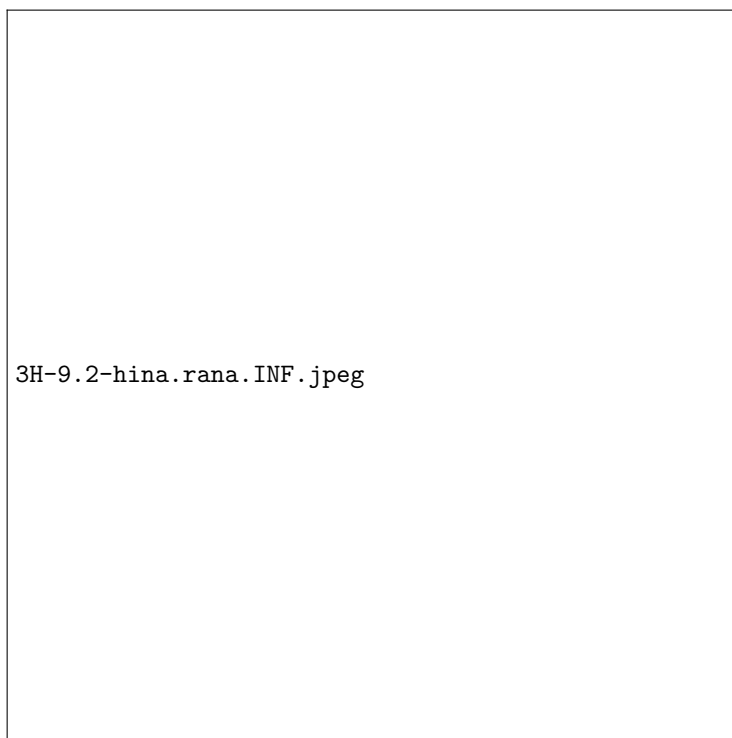


Figure 1: Grafiek van $r = 2 - \cos \theta$